

Multinomial Distribution

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Graded Assignment • 10 min

**Due** May 14, 11:59 PM PDT

^ Overview

In this assignment, you will consider the Multinomial Distribution as a K dimensional expansion of the Bernoulli distribution's binary random variable. There are a few reflection questions before the exercises in this section. In the following video, Peter will discuss the maximum likelihood estimate for the Multinomial distribution.

Make sure that you look at the feedback provided after you submit this assignment, even if you correctly answered every question.

^ One in K Variables

In contrast to the random variables we have explored thus far, which have been binary random variables, we now consider a case where one output out of K possible classes is selected. The random variable x has only one element where $x_k = 1$, and all of the other elements are 0.

For example, when you roll a 6-sided die, you take whichever number is face up. Possible values you can roll range from 1 to 6. If we rolled a 3, then,

$$\vec{x} = (0, 0, 1, 0, 0, 0)^T.$$

If we sum the x vector, it should always be equal to 1 such that $\sum_{k=1}^K x_k = 1$.

^ The Categorical Distribution

Each potential class, x_k has an associated probability of occurring, μ_k . Each μ_k must be between 0 and 1 since it describes a probability, and the sum of μ_k s should sum to 1, such that $0 \leq \mu_k \leq 1$ and $\sum_{k=1}^K \mu_k = 1$.